

Response to Reviewers
Manuscript ToG-2026-0064:
“From Rules to Nash Equilibria: Formally Verified Game-Theoretic
Analysis of a Competitive Trading Card Game”

We thank the Editor-in-Chief, the Associate Editor, and the three reviewers for their careful and constructive reading. The reviewers found the work novel, technically impressive, and a valuable application of formal verification to game analysis; their main concern was that several conclusions were stated more strongly than the modeling assumptions justify, together with requests for greater accessibility, robustness reporting, and clarity about verification boundaries. We have revised the manuscript thoroughly to address every point.

How to read the revision. All substantive additions and changes are typeset in [blue](#) in the revised manuscript (a macro `\highlightchangesfalse` produces a clean camera-ready version). Section references below point to the revised PDF.

Summary of the most important changes.

1. **Correctness fix to the symmetric-equilibrium numbers.** While responding to Reviewer 3 we discovered that the symmetric-Nash gap figures reported in the previous version (Dragapult “416.7%, gap 63.3,” “five best responses”) were inherited from an early, superseded enumeration of a *different* game model and did not match the Lean artifact that the paper actually verifies (`SymmetricNash.lean`). The verified symmetric equilibrium has value exactly 500, *seven*-deck support, and places Dragapult at 459.6%—a strict gap of **40.4%**. We recomputed every affected number with exact rational arithmetic, cross-checked it against the Lean theorems, and corrected the abstract, Table VII (and its caption), and Section VII. We are grateful that the review process surfaced this; the corrected figures are now fully consistent with the machine-checked artifact.
2. **Dragapult’s exclusion is now shown to be equilibrium-independent** (Reviewer 3, Major #1–2): a complementary-slackness argument establishes that a strict best-response gap against *one* optimal strategy excludes a deck from the support of *every* equilibrium—symmetric or asymmetric.
3. **New accessibility material** (Reviewer 2): a Lean/formal-verification primer (Sec. III-A), a notation table (Table I), a verification-pipeline diagram (Fig. 1), and uniform captions on all code listings.
4. **Foregrounded qualifications** (Reviewer 1): “suboptimal” is now explicitly model-relative in the abstract, Sec. X, and the conclusion, and the paper is framed as a methodological case study.
5. **Title reframed as a case study** (Reviewer 1, AE): the title now reads “...A Lean 4 Case Study in Game-Theoretic Analysis...” instead of “Formally Verified...,” aligning the headline framing with the more measured, case-study presentation requested throughout.

6. **Artifact counts reconciled exactly with the source** (Reviewer 2). We re-audited the published artifact and corrected every count to its exact, grep-reproducible value: **87 files**, **~31,900 lines**, **2,627 theorems**, of which **145** use `native_decide` and **2,482** are kernel-checked (Table IX rebuilt with the true per-category split; previous version’s category counts were stale). One displayed expected-win-rate value (Ceruledge) was corrected from 44.8% to its exact 45.4%. These were the only numeric discrepancies we found on a full recheck.
7. **The equilibrium-independent exclusion is now machine-backed and global** (Reviewer 3). We added a Lean theorem (`symmetric_nash_dragapult_strict_gap`) certifying Dragapult’s strict sub-value payoff, and a linear program (`dragapult_universal_exclusion.py`) that maximizes Dragapult’s payoff over the *entire* optimal-strategy set ($459.6 < 500$), upgrading the earlier size-bounded search to a global certificate. We also replaced a stale supplementary results file that reported the superseded ($A=M, B=M^\top$) model with a corrected enumeration (`enumerate_symmetric_nash.py`) consistent with the paper.

Associate Editor

“some conclusions are overstated relative to the underlying modeling assumptions ... the interpretation of equilibrium results and claims regarding strategy optimality require more careful qualification ... dependence on subjective modeling choices, the limited empirical scope, and the need for clearer discussion of verification boundaries, robustness, and supporting artifacts ... improved accessibility.”

Response. We have acted on each item: (i) equilibrium claims are re-qualified and the key exclusion result is now proven equilibrium-independent (Sec. VII); (ii) “optimality/suboptimality” is consistently scoped to the single-match payoff model (abstract, Sec. X, conclusion); (iii) the dependence on subjective type assignments is tested and bounded (Sec. III-C); (iv) empirical scope is reframed as a case study (Sec. X, conclusion); (v) the verification boundary is classified explicitly as an engineering bottleneck rather than a fundamental limit (Sec. IX-A); (vi) robustness is reported via explicit perturbation (ε) bounds (Sec. X); (vii) accessibility is improved with a primer, notation table, and pipeline diagram. Details follow per reviewer.

Reviewer 1

(1) The paper leans heavily on “formally verified”/“machine-checked” in a way that may overstate assurance, given the reliance on `native_decide`.

Response. We added a measured framing throughout, beginning with the **title**, which now reads “...A Lean 4 Case Study...” rather than “Formally Verified...” The abstract retains the explicit trust-boundary sentence and now adds the equilibrium-independence qualifier; the assurance gradient is summarized in Table IX (kernel- vs. compiler-checked, with exact counts: 2,482 kernel-checked vs. 145 `native_decide`), and Sec. IX-A states plainly that `native_decide` produces no kernel-checked proof term. We also classify a fully kernel-checked version as a tractable engineering task (see Reviewer 3, Minor #3).

(2) The bridge from rules to empirical conclusions depends on subjective type assignments; this should be tested more carefully.

Response. Sec. III-C now separates load-bearing from borderline assignments and adds a *quantitative* sensitivity test. Only genuine multi-type decks are borderline; the single assignment the popularity paradox depends on—Dragapult Dusknair as a Psychic defender, weak to the unambiguously Dark counters Grimmsnarl, Mega Absol, and N’s Zoroark—is *not* borderline. Enumerating every defensible reassignment of the borderline decks places the aggregate alignment rate in the range [70.6%, 87.5%] (point estimate 83.3%), all far above the 50% random baseline, while the headline result is invariant across all of them. We now report this range explicitly and label the alignment figure as a descriptive, sensitivity-bearing statistic rather than a verified claim.

(3) *The equilibrium narration is too confident; the exact support is fragile, and conclusions should distinguish the point estimate from what is stable under perturbation.*

Response. We now make this distinction central. Sec. VII and the new “Equilibrium robustness” paragraph (Sec. X) separate (a) the *perturbation-stable* exclusion of Dragapult, which tolerates an entrywise perturbation of $\approx 20\%$, from (b) the *point-estimate-specific* support composition, whose marginal members flip under $\approx 5\%$. Our headline conclusion rests only on (a).

(4) *“Suboptimal” needs cautious handling; the bimatrix/Swiss/consistency/familiarity caveat should be brought to the foreground, especially the abstract and conclusion.*

Response. Done. The abstract now states that “suboptimal” is relative to a single-match payoff model omitting Swiss and consistency incentives; the conclusion repeats this; and Sec. X (“Strategic objective mismatch”) spells out that a player may rationally choose an in-model-suboptimal deck for Swiss consistency, familiarity/misplay-risk, event stability, or card access.

(5) *The empirical scope is narrow (three-week window, top-14, third-party pipeline); present the work as a methodological case study.*

Response. We adopt this framing explicitly. The abstract calls the contribution “a reproducible case study”; Sec. X closes by reading the contribution as a methodological case study for one well-instrumented format and window rather than a universal metagame claim; data-provenance biases remain itemized in Sec. V-C.

Reviewer 2 (Minor Revision)

Accessibility and artifact

The manuscript assumes substantial prior knowledge of formal verification; the LEAN source was not included in the submission package.

Response. We added a dedicated primer (Sec. III-A) introducing proof assistants, Lean 4, decision procedures, and exact arithmetic for the general AI/PCG audience, and a pipeline diagram (Fig. 1). On the artifact: the complete anonymized Lean source (**87 files, $\sim 31,900$ lines, 2,627 theorems, no sorry/admit/axiom**—confirmed by `#print axioms` on the headline theorems, which depend only on `propext/Quot.sound` and the `native_decide` axiom `Lean.ofReduceBool`), the 14×14 matrix, and the Python scripts are provided as supplementary material and re-verify end-to-end via a single `lake build`. We re-audited every count against the source so the manuscript and artifact now agree exactly, removed a superseded results file and the raw-model enumeration scripts it came from, and added two reproduction scripts (`enumerate_symmetric_nash.py`, `dragapult_universal_exclusion.py`). The strengthened Data Availability statement maps each numerical claim to a named theorem so any result can be located and re-run. The archive is attached to this resubmission.

Clarifications

1. (V-A) *Rationale for weighting a tie as $T/3$.*

Response. Added (Sec. V-A): the coefficient mirrors the Pokémon Championship Series match-point system (win = 3, tie = 1, loss = 0), so a tie is worth exactly one-third of a win in the format’s *own* reward structure—encoded in the artifact as `scoreWin=3`, `scoreDraw=1`, `scoreLoss=0`. This is more faithful than the common $T/2$ convention. We also note the symmetric-equilibrium support is invariant to this coefficient (Sec. X).

2. (IX-B, XI-A) *What does the high “fixed cost” of porting entail?*

Response. Sec. IX-B now itemizes it: (i) encoding the strategic-layer rule semantics (type chart, legality, prize/turn structure)—the bulk of the code and the only part needing real domain modeling; (ii) defining the archetype enumeration and matrix ingestion; (iii) reusing the game-theory harness unchanged, since it is parameterized over an abstract `FiniteGame`. Steps (ii)–(iii) are days of work given a matrix; step (i) dominates and scales with rule complexity. A fresh archetype set within the same game is the ~ 200 -line marginal case.

3. (VIII) *How does the Swiss objective specifically influence the analysis?*

Response. Sec. VIII now explains that our Nash/replicator results optimize a single-match criterion, whereas Swiss qualification optimizes a *threshold* functional $P(X-2)$ over eight rounds. Because the threshold is sensitive to outcome *variance*, not only the mean, it re-weights the single-match equilibrium toward consistent decks for strong contenders near the cut-line, which can rationalize a portion of the observed “suboptimal” play. We mark a full Swiss-utility equilibrium as future work.

4. (II) *Wilson intervals and perturbations make the method a hybrid, complementary to Monte Carlo, rather than orthogonal.*

Response. We agree and rephrased Sec. II accordingly: the approach is now characterized as a *hybrid* that applies formal verification *on top of* statistical estimation—Wilson intervals for per-cell uncertainty and a 10,000-iteration Monte Carlo resampling for equilibrium robustness—with Lean certifying exact claims over each sampled or point-estimate matrix. Fig. 1 visualizes this interface.

Suggestions

1. *Inconsistent listing captions.*

Response. All code listings now carry captions (Listings 1–8).

2. *A notation table would help.*

Response. Added (Table I), covering s_j , $w_{i,j}$, $\mathbb{E}[\text{WR}_i]$, M , S_{ij} , the game value v , replicator x_i , f_i , $\bar{f}$, the Wilson $\tilde{p}$, P_{Bo3} , the Swiss cut-line, and the permil scale.

3. *A brief intro to LEAN/formal verification.*

Response. Added as Sec. III-A (see above).

4. *A pipeline diagram clarifying the Python $\leftrightarrow$ LEAN interface and compile-time vs. runtime.*

Response. Added as Fig. 1: an untrusted Python discovery layer only *proposes* candidates; Lean *certifies* every claim at compile time over exact rationals; a separate runtime stage executes demonstration traces.

Reviewer 3 (Major Revision)

Major #1. Uniqueness via symmetric equilibria is a conceptual leap: asymmetric equilibria (A, B) are interchangeable and equally rational, so uniqueness is an artifact of restricting to symmetric profiles, not a consequence of individual rationality.

Response. We agree with the theory and have rewritten Sec. VII to make the stronger, *equilibrium-independent* argument. We now explicitly acknowledge interchangeability and that symmetry is a modeling choice. We then show that Dragapult’s exclusion does *not* rely on the symmetry restriction: in a constant-sum game, complementary slackness forces the support of *every* optimal strategy to lie inside the best-response set of *every* optimal opponent strategy. Since the symmetric strategy is optimal for both players and Dragapult earns 40.4% *strictly* below the value against it, Dragapult is a best response to no optimal strategy and therefore appears in *no* equilibrium—symmetric or asymmetric. This strict gap is now itself machine-checked (`symmetric_nash_dragapult_strict_gap`), and we corroborate the universal claim with a *global* linear program (`dragapult_universal_exclusion.py`) that maximizes Dragapult’s payoff over the *entire* optimal-strategy set and returns $459.6 < 500$ —a stronger certificate than the earlier size-bounded support search. We now state that the symmetry restriction only selects a single *canonical* profile to report, whereas the *exclusion* is a strict best-response fact holding across the entire equilibrium set; we also delimit the scope precisely (the universal claim is exact for the constant-sum symmetrization, with the approximately-constant-sum raw game corroborating via its directly verified zero-weight equilibria).

Major #2. Chance timing/evolutionary interpretation: under an asymmetric equilibrium the population would split into two sub-populations; the manuscript implicitly shifts from classical to bounded/evolutionary rationality and should decouple the two and state the epistemic boundary of the single-population replicator.

Response. Added a dedicated paragraph, “Classical versus evolutionary rationality” (Sec. VII-A). We distinguish the classical one-shot equilibrium question from the evolutionary imitation dynamics, note that an asymmetric profile (A, B) would physically split the population into two behavioral sub-populations, and state that the single-population replicator is valid for *macro* share dynamics only under a specific epistemic regime: an approximately homogeneous population, largely unstructured matchmaking, and open centralized information flow (net-decking). We emphasize this is an assumption, not a theorem, and that the Dragapult-exclusion result lives at the static-equilibrium level and does not depend on the evolutionary reading.

Major #3. Quantal response/noise: the proof rests on a knife-edge, noise-free matrix; under payoff perturbations asymmetric equilibria evaporate while the symmetric profile is a robust focal point. Discuss how the symmetric equilibrium withstands micro-level payoff uncertainty (Sec. VI-A).

Response. Added a paragraph to Sec. VI-A. We make two points. First, the load-bearing claim is *not* knife-edge: Dragapult’s 40.4% gap tolerates $\approx 20\%$ of entrywise perturbation, several times the $\pm 1.8\text{--}9\%$ Wilson sampling error. Second, noise is precisely why we anchor on the symmetric profile: asymmetric equilibria of a near-constant-sum game are highly sensitive to matrix drift, whereas the symmetric (maximin) strategy is the self-correcting focal point that stochastic, quantal-response play converges toward; the sensitivity analysis ($> 96\%$ core inclusion) is its empirical signature.

Minor #1. The `GrimssnarlFroslas` identifier typo; state explicitly that it is a benign string-matching artifact.

Response. The footnote now states explicitly that the identifier is a purely cosmetic string-matching artifact used consistently throughout, with no effect on any definition, theorem, or result, retained verbatim for traceability.

Minor #2. State the exact ε bound under which the symmetric Nash support set remains invariant.

Response. Added (Sec. X, “Equilibrium robustness”). A uniform entrywise perturbation of size ε moves any payoff-vs-mix and the value by at most ε , so a deck with best-response gap g cannot enter the support while $\varepsilon < g/2$. The smallest exclusion gap among non-support decks is 10.3‰, so an excluded deck can enter only once $\varepsilon \gtrsim 5\%$; we now also note the opposite direction (a thin incumbent such as Alakazam, at 2.5% weight, can exit under a comparable perturbation), so the single ε guaranteeing the *entire* support is $\approx 5\%$. Dragapult’s exclusion, with gap 40.4‰, is by contrast invariant for $\varepsilon \lesssim 20\%$ (a four-fold larger margin). For the tie coefficient specifically the bound is exact: because each matchup and its mirror share one record, the tie terms cancel in $M_{ij} - M_{ji}$, so the symmetrization is algebraically independent of the tie weight; we verified $\max_{ij} |S_{ij}^{T/3} - S_{ij}^{T/2}| < 1\%$ and that re-enumeration under $T/2$ reproduces the identical seven-deck support.

Minor #3. State whether a kernel-checked alternative to `native_decide` is fundamentally impossible or merely an engineering bottleneck.

Response. Sec. IX-A now classifies it explicitly as an *engineering bottleneck*, *not a fundamental impossibility*: the obstruction is the kernel’s inefficient reduction of `Fin.foldl`, and a kernel-transparent reimplementation (e.g., structurally recursive predicates or inductive relations instead of opaque folds) is possible in principle at the cost of substantial proof-engineering effort and longer kernel-reduction times.

IEEE AI-content policy

Per the Editor’s note, we added a “Disclosure of Generative AI Use” statement: generative AI assistance was used during revision for copy-editing/reorganization and for cross-checking numerical computations; all formal results are machine-checked in Lean 4 and were independently verified by the authors, who take full responsibility for the content.

We believe these revisions resolve the reviewers’ concerns while preserving the artifact’s rigor, and we thank the reviewers again for feedback that materially strengthened both the paper and its correctness.